

An Analytical Integrator for Solid-Fluid Coupled Buoyancy Forces

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Abstract

Accurate computation of buoyancy forces in fluid-rigid coupling involves integrating nonlinear functions over complex surfaces, which requires careful handling. We propose a closed-form integrator tailored for triangular meshes that eliminates discretization-dependent errors, enabling exact computation of net forces and torques based on Archimedes' principle. Our method accurately matches analytical solutions reported in the fluid engineering literature, where Gaussian quadrature fails to achieve comparable precision. We further demonstrate that our integrator produces stable animations and can be applied as a post-process to fluid surfaces that approximate hydrostatic pressure conditions, ensuring broad compatibility with height-field fluids, grid-based and surface-based simulation techniques, as well as artist-generated surfaces.

CCS Concepts

• Computing methodologies → Modeling and simulation.

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1 Introduction

Height field representations are able to express horizontally expanding water surfaces, such as oceans and lakes, and have the potential to enable efficient coupling with rigid body dynamics [Bridson 2015]. This coupling presents significant challenges, particularly in accurate buoyancy computation. For instance, when floating objects such as boats or debris interact with surrounding water, accurately computing buoyancy forces is crucial for maintaining momentum conservation in physically plausible animations.

To this end, Gaussian quadrature rules may be employed to integrate forces and torques induced by buoyancy. However, their accuracy is highly sensitive to mesh discretization [Alauzet and Loseille 2016] because the underlying torques are not polynomial functions of depth, despite the magnitude of pressure forces being linear in depth.

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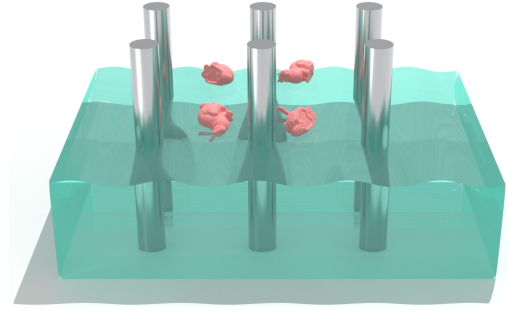


Figure 1: Real-time fluid-rigid body coupling demonstrating a floating bunny on dynamic fluid surfaces. Our closed-form buoyancy integration method enables stable, mesh-independent floating behavior with broad compatibility across fluid surface representations, while maintaining computational efficiency suitable for real-time applications.

In particular, we observe that significant variations in mesh resolution and geometric distortions exacerbate numerical errors, especially violating angular momentum conservation, which leads to the generation of ghost torques.

This problem has been long recognized in Computational Fluid Dynamics (CFD), where higher accuracy is typically achieved through increased mesh resolution [Alauzet and Loseille 2016; Zheleznyakova 2020], albeit at significant computational cost [Bajo et al. 2020].

In graphics, buoyancy has been accurately modeled using volumetric discretization [Batty et al. 2007; Bridson 2015]; however, height field approaches have primarily considered only translational motion [Chentanez and Müller 2010; Wang et al. 2007]. We note that although rigid body rotation appears in these animations, the underlying mechanism (particularly the computation of torque) is not explicitly described.

In this paper, we propose a novel closed-form integrator that computes both net forces and torques per triangle exactly, up to floating-point precision. Additionally, we introduce a new angular velocity damping method that operates by scaling angular momentum, thereby mitigating artifacts that arise from direct angular velocity damping. Our contributions are summarized as follows:

- Physics-driven rotational buoyancy coupling compatible with fluid simulations that reasonably approximate hydrostatic pressure conditions.
- A closed-form integrator for computing buoyancy-induced forces and torques per triangle, accurate up to numerical precision.
- A novel angular velocity damping method based on angular momentum scaling.

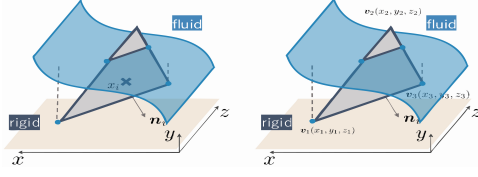


Figure 2: Comparative overview of buoyancy force computation approaches. Left: Conventional numerical integration using single-point quadrature [Zheleznyakova 2020], which requires high mesh resolution for accuracy and suffers from discretization artifacts. Right: Our proposed closed-form algebraic integration method, which achieves mesh-independent accuracy through symbolic computation of volume integrals, eliminating discretization errors while significantly reducing computational cost.

2 Related Work

Solid–Fluid Coupling for Height Fields. Early approaches by [O’Brien and Hodgins 1995; Wang et al. 2007] modeled vertical forces based on submerged depth, bypassing explicit computation of distributed fluid pressure or buoyancy. [Chentanez and Müller 2010] later introduced a two-way coupling scheme between fluid and rigid bodies, employing an explicit formulation on triangular meshes. However, neither method describes torque computation, leaving the mechanism by which rotational motion is achieved unclear.

Volumetric Coupling. [Carlson et al. 2004] introduced a unified framework treating the velocity field of a rigid body as a seamless boundary condition. While this approach works in practice, it requires special care near rigid-body interfaces to handle discontinuities, making it potentially prone to numerical errors. Subsequent work by [Batty et al. 2007] introduced a variational formulation based on volumetric integration; however, such approaches require full volumetric fluid simulation and are not directly applicable to height field-based coupling methods.

GPU-Accelerated Method. [Bajo et al. 2020] introduced a GPU-accelerated technique leveraging Archimedes’ principle, using surface parameterization to store both normal and area information in textures, enabling support for heterogeneous materials and real-time applications. However, mesh parameterization is non-trivial and requires dedicated pre-computation.

Direct Integrations on Meshes. [Weißmann and Pinkall 2012] demonstrated that analytical formulations exist for submerged rigid bodies using Kirchhoff tensors, without requiring full fluid simulation of the surrounding medium. This grid-free approach enables fast computation of buoyancy and added mass effects, though it is demonstrated only for fully submerged rigid bodies. [Zheleznyakova 2020] applied mesh-based numerical integration using single-point quadrature to couple floating rigid bodies with dynamic wave surfaces. While computationally efficient, their method is ultimately limited by mesh quality due to its reliance on Gaussian quadrature over surface elements.

3 Our Closed-Form Integrator

Figure 2 provides an overview of our proposed algebraic integration method compared to the single-point quadrature integration approach. We start from Archimedes’ principle, which states that the buoyancy force is equal in magnitude and opposite in direction to the gravitational force acting on the submerged volume of the rigid body. This can be written as

$$F_{\text{buoyancy}} = -\rho V g, \quad (1)$$

where F_{buoyancy} denotes the buoyancy force, V is the submerged volume, and g is the gravitational acceleration vector. Note that this formulation provides only net force, not torque. To compute torque, a detailed analysis of pressure distribution is required. Alternatively,

$$F_{\text{buoyancy}} = -\rho V g = -\iiint_V \rho g dV = -\iiint_V \nabla p dV = -\iint_S p n dS, \quad (2)$$

where $p = \rho g(h - y)$ denotes the hydrostatic pressure at vertical position y . This confirms that our approach integrates naturally with fluid surfaces that reasonably approximate hydrostatic pressure distributions. Substituting the pressure expression, we obtain

$$F_{\text{buoyancy}} = -\iint_S \rho g(h - y) n dS. \quad (3)$$

Using the above relation, the torque can be similarly expressed as

$$T = -\iint_S (x - x_{\text{centroid}}) \times \rho g(h - y) n dS, \quad (4)$$

where S and x_{centroid} represent the submerged surface of the rigid body and its center of mass, respectively. On triangle meshes, (3) and (4) can be symbolically expanded as

$$F_{\text{buoyancy}} = \sum_i \frac{\rho g S_i}{3} (-3h_i + y_1 + y_2 + y_3) n_i, \quad (5)$$

$$T = \sum_i \frac{\rho g S_i}{12} [\{12h_i - 4(y_1 + y_2 + y_3)\} x_{\text{centroid}} + A] \times n_i, \quad (6)$$

where the auxiliary vector A has the following three components:

$$A_1 = (x_1 + x_2 + x_3) \{-4h_i + 2(y_1 + y_2 + y_3)\} + x_1 y_1 + x_2 y_2 + x_3 y_3, \quad (7)$$

$$A_2 = (y_1 + y_2 + y_3) \{-4h_i + 2(y_1 + y_2 + y_3)\} - 2(y_1 y_2 + y_2 y_3 + y_3 y_1), \quad (8)$$

$$A_3 = (z_1 + z_2 + z_3) \{-4h_i + 2(y_1 + y_2 + y_3)\} + z_1 y_1 + z_2 y_2 + z_3 y_3, \quad (9)$$

where $x_1, y_1, z_1, x_2, y_2, z_2$, and x_3, y_3, z_3 are the xyz-coordinates of each triangle’s three vertices, n_i is its outward normal, S_i is its area, and h_i is the fluid surface elevation at the triangle’s center of mass.

By substituting the mesh vertex coordinates, normals, and water heights, we obtain closed-form expressions for buoyancy forces and torques. These forces enable one-way coupling from compatible fluid surface representations to rigid bodies and are integrated over time using forward Euler to simulate rigid body motion.

3.1 Triangle Clipping with Water Surface

We use a marching triangles to clip triangles against the water surface $y = h(x, z)$. For each triangle vertex v_j , we evaluate $d_j = h(v_j^x, v_j^z) - v_j^y$. Triangles with mixed signs are clipped using linear interpolation along intersecting edges to generate submerged triangles for accurate force integration. When clipping generates quadrilateral surfaces, they are triangulated on the fly.

Algorithm 1 Our Analytical Computation Algorithm

Require: Rigid body mesh, height field function $h(x, z)$, gravity g , fluid density ρ

Ensure: Total buoyancy force F_b and torque T_b

```

1:  $\mathbf{x}_{\text{com}} \leftarrow$  center of mass of rigid body
2:  $F_b \leftarrow \mathbf{0}$ ,  $T_b \leftarrow \mathbf{0}$ 
3: for each triangle  $T_i$  with vertices  $(v_0, v_1, v_2)$  do
4:    $d_j \leftarrow h(v_j^x, v_j^z) - v_j^y$  for  $j = 0, 1, 2$ 
5:   if any  $d_j > 0$  then
6:      $\mathcal{T}_{\text{sub}} \leftarrow$  submerged triangle(s)
7:     for each triangle  $t \in \mathcal{T}_{\text{sub}}$  do
8:        $f_b \leftarrow$  compute buoyancy force using (5)
9:        $t_b \leftarrow$  compute torque using (6) about  $\mathbf{x}_{\text{com}}$ 
10:       $F_b, T_b \leftarrow F_b + f_b, T_b + t_b$ 
11:     end for
12:   end if
13: end for
14: return  $F_b, T_b$ 

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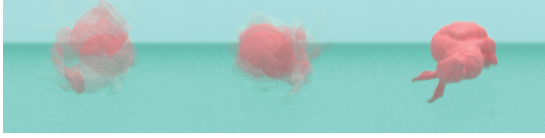


Figure 3: Comparative analysis of rotational damping strategies for floating objects on static fluid surfaces. Left: No damping applied, showing persistent oscillatory motion. Center: Damping applied directly to angular velocity, resulting in minimal decay and visual artifacts. Right: Damping applied to angular momentum (our approach), demonstrating effective rotational stabilization. Each visualization overlays frames 340-400 at 60 FPS with 10-frame intervals and transparency to illustrate motion decay patterns. Only angular momentum damping achieves physically plausible stabilization without artifacts.

4 Damping Angular Motion

Damping stabilizes rigid body motion in floating object simulations [Egan 2003]. Unlike translational damping, rotational damping requires applying to angular momentum rather than angular velocity to avoid visual artifacts. The rotational update follows: (i) calculate torque T , (ii) update angular momentum as $\frac{dL}{dt} = T$, (iii) apply damping $L_{\text{damped}} = \alpha \cdot L$ and compute $\omega = I^{-1}L_{\text{damped}}$, (iv) update quaternion as $\frac{\partial q}{\partial t} = \frac{1}{2}\omega q$. We apply damping coefficient α (where $0 \leq \alpha \leq 1$) to scale angular momentum before computing angular velocity. This approach produces visually and numerically consistent results compared to damping angular velocity directly, as demonstrated in Figure 3 and 4.

5 Validations

Experimental Setup. We evaluate the proposed method against the single-point quadrature numerical integration scheme [Zheleznyakova 2020], focusing on accuracy, stability, and computational efficiency by comparing computed results with analytical solutions [Igarashi

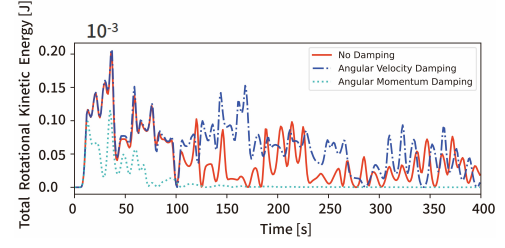


Figure 4: Analysis of the scene shown in Figure 3: Comparison of total rotational kinetic energy for the damping simulations. Our angular-momentum damping approach achieves the greatest rotational stabilization over time.

and Nakamura 2007]. While higher-order quadrature schemes may improve accuracy, they alone do not completely eliminate discretization-induced errors because the torque is not polynomial. We prepared two types of cube meshes with varying resolutions: regular meshes (24, 96, 384, 1,536, and 6,144 elements) and irregular meshes (12, 48, 192, 768, 3,072, and 12,288 elements). For each mesh, the initial orientation angle θ was set to the analytically known equilibrium angle of 26.565° [Igarashi and Nakamura 2007], at which the net buoyancy torque theoretically counteracts and vanishes.

To isolate the effects of buoyancy and torque calculations, we set the fluid surface height field to be constant across the domain. The density ratio between the fluid and rigid body was set to $1 : 0.75$, ensuring that the center of gravity remains above the center of buoyancy, thereby promoting rapid convergence to static equilibrium. The time step was fixed at $\Delta t = \frac{1}{60}$ s, corresponding to 60 FPS, as is typical in real-time applications.

5.1 Accuracy, Stability, and Computational Performance

We conducted experiments to assess whether the cube maintains a stationary posture at the equilibrium angle, thereby evaluating solution stability and computational performance.

We verified that the buoyancy torque approaches zero and that results are independent of mesh resolution and quality. When the buoyancy torque falls below 10^{-3} relative to the inertia tensor scale (approximately 10^{-4} in our setting), the resulting angular acceleration becomes negligible, indicating rotational equilibrium. In practical terms, a torque magnitude of 10^{-7} or less is deemed sufficiently accurate.

Visual results are presented in Figure 5, while computed torque values are summarized in Tables 1 and 2.

Single-point quadrature requires at least 1,536 elements for regular meshes and 12,288 elements for irregular meshes to achieve acceptable torque accuracy. For single-point quadrature, irregular meshes require approximately $8\times$ the resolution of regular meshes to achieve stable results. In contrast, our proposed method achieves comparable accuracy with only 24 regular or 12 irregular elements, representing the minimal resolution required for accurate torque computation. Our method consistently produces stable behavior at the equilibrium angle [Igarashi and Nakamura 2007], independent of mesh resolution or quality.

Table 1: Torque Calculation Results for Regular Meshes at Equilibrium Angle θ

Elements	Zheleznyakova et al.			Our Method		
	$T_x [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_y [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_z [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_x [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_y [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_z [\text{kg} \cdot \text{m}^2/\text{s}^2]$
24	-2.17×10^{-19}	-2.17×10^{-19}	-2.38×10^{-4}	-3.47×10^{-18}	8.67×10^{-19}	-3.00×10^{-9}
96	0.00	0.00	-2.97×10^{-5}	-4.34×10^{-19}	0.00	-1.36×10^{-20}
384	2.20×10^{-19}	9.49×10^{-20}	-3.72×10^{-5}	1.46×10^{-19}	-1.15×10^{-19}	6.68×10^{-9}
1536	0.00	4.69×10^{-19}	-4.67×10^{-7}	4.57×10^{-19}	5.48×10^{-19}	-1.26×10^{-9}
6144	1.71×10^{-19}	2.87×10^{-19}	-5.45×10^{-8}	-3.76×10^{-19}	2.84×10^{-19}	1.90×10^{-9}

Table 2: Torque Calculation Results for Irregular Meshes at Equilibrium Angle θ

Elements	Zheleznyakova et al.			Our Method		
	$T_x [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_y [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_z [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_x [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_y [\text{kg} \cdot \text{m}^2/\text{s}^2]$	$T_z [\text{kg} \cdot \text{m}^2/\text{s}^2]$
12	7.80×10^{-4}	-4.97×10^{-4}	-6.34×10^{-4}	0.00	0.00	-1.09×10^{-9}
48	1.60×10^{-4}	-5.32×10^{-5}	-2.00×10^{-9}	-8.30×10^{-9}	1.70×10^{-9}	1.40×10^{-8}
192	3.99×10^{-5}	-1.33×10^{-5}	1.04×10^{-9}	-1.00×10^{-11}	2.68×10^{-9}	4.40×10^{-9}
768	9.98×10^{-6}	-3.33×10^{-6}	-6.92×10^{-9}	9.22×10^{-10}	-1.50×10^{-9}	-9.20×10^{-10}
3072	2.49×10^{-6}	-8.31×10^{-7}	1.61×10^{-9}	-3.28×10^{-10}	-7.39×10^{-10}	2.67×10^{-9}
12288	6.24×10^{-7}	-2.07×10^{-7}	1.55×10^{-9}	1.06×10^{-10}	1.17×10^{-10}	3.58×10^{-9}

In terms of computational performance, single-point quadrature incurs costs of 7.14×10^{-4} s/frame for regular meshes with 1,536 triangular elements and 5.19×10^{-3} s/frame for irregular meshes with 12,288 triangular elements to achieve sufficient accuracy. Our proposed method achieves comparable accuracy with runtimes of only 3.64×10^{-5} s/frame for regular meshes with 24 elements and 2.52×10^{-5} s/frame for irregular meshes with 12 elements. This corresponds to speedups of $19.6\times$ for regular meshes and $206\times$ for irregular meshes. These results confirm that the proposed method not only achieves mesh-independent accuracy but also significantly reduces computational costs, making it particularly advantageous for real-time fluid–rigid body coupling applications and compatible with both simulated and artist-generated fluid surfaces.

Figure 1 demonstrates the successful integration of our method with analytically generated fluid surfaces using trigonometric functions, showing stable floating behavior that exemplifies the method’s compatibility with non-simulated surface representations. This demonstrates that our approach is robust to mesh quality and highly suitable for real-time applications.

6 Conclusion

This paper presented a novel closed-form method for computing buoyancy forces and torques that is independent of rigid body mesh resolution and quality. Our method achieves comparable accuracy to single-point quadrature using significantly fewer elements: 24 regular versus 1,536, and 12 irregular versus 12,288.

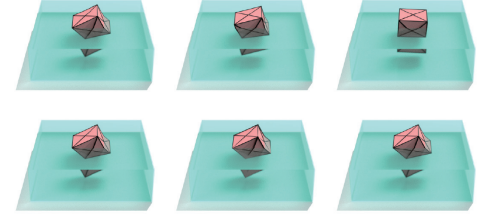
This yields speedups of $19.6\times$ for regular meshes and $206\times$ for irregular meshes when single-point quadrature uses high resolutions to match our accuracy. Future work will extend the method to fully two-way coupled frameworks for comprehensive fluid–structure interaction.

Acknowledgments

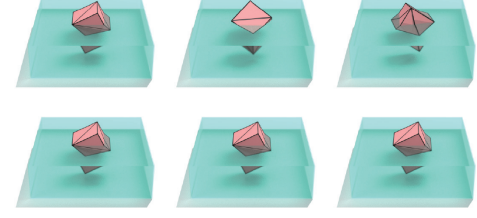
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(a) Regular mesh (24 elements).



(b) Irregular mesh (12 elements).

Figure 5: Quantitative validation of mesh-independent accuracy in buoyancy force computation. Top: Regular mesh (24 elements) and Bottom: Irregular mesh (12 elements). For each configuration, the upper panel shows results using conventional numerical integration [Zheleznyakova 2020], while the lower panel demonstrates our closed-form algebraic integration. Temporal progression shows frames 0, 50, and 100 (left to right). Our method achieves stable equilibrium behavior independent of mesh quality, while numerical integration exhibits mesh-dependent artifacts and instability, particularly evident in the irregular mesh case.

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